



PHYSICAL AND STRUCTURAL GEOLOGY

GS 21/L

Historical Geology



The Importance of Time Scale

Interpreting Earth history is a prime goal of the science of geology. By studying rocks, especially sedimentary rocks, and the features they contain, geologists can unravel the complexities of the past. Among geology's major contributions to human knowledge are the geologic time scale and the discovery that Earth history is exceedingly long.



Two dating techniques:

1. *Relative dating* - such dates cannot tell us how long ago something took place, only that it followed one event and preceded another. It is done by placing the rocks in their respective order of formation. Non-numerical in nature.
2. *Absolute dating or Numerical dating* - such dates specify the actual number of years that have passed since an event occurred.

Relative Dating

Basic Geologic Principles

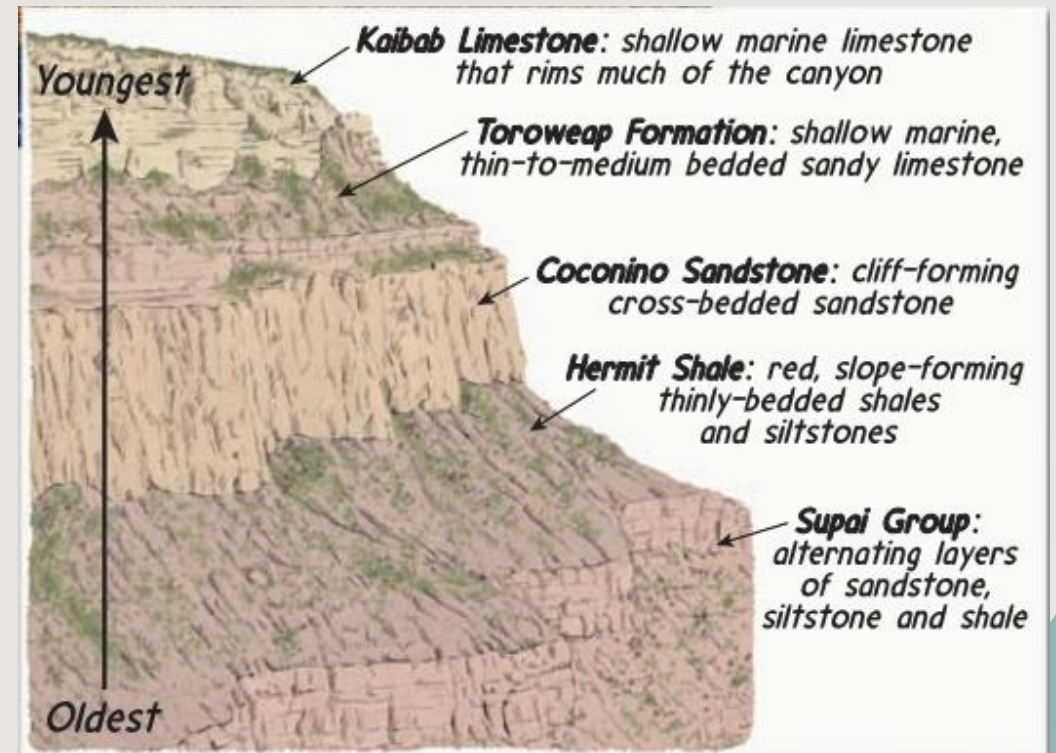
1. Principle of Superposition
2. Principle of Original Horizontality
3. Principle of Lateral Continuity
4. Principle of Cross-cutting Relationships
5. Inclusions
6. Unconformity

Basic Geologic Principles

1. *Principle of Superposition* - the law simply states that in an undeformed sequence of sedimentary rocks, each bed is older than the one above and younger than the one below.



Figure 4.1 Superposition The Grand Canyon, the Supai Group is oldest and the Kaibab Limestone is youngest.



Basic Geologic Principles

2. *Principle of Original Horizontality* - layers of sediment are generally deposited in a horizontal position.

Figure 4.2 Original horizontality

Most layers of sediment are deposited in a nearly horizontal position. When we see strata that are folded or tilted, we can assume that they were moved into that position by crustal disturbances after their deposition. (Photo by Marco Simoni/Robert Harding World Imagery)

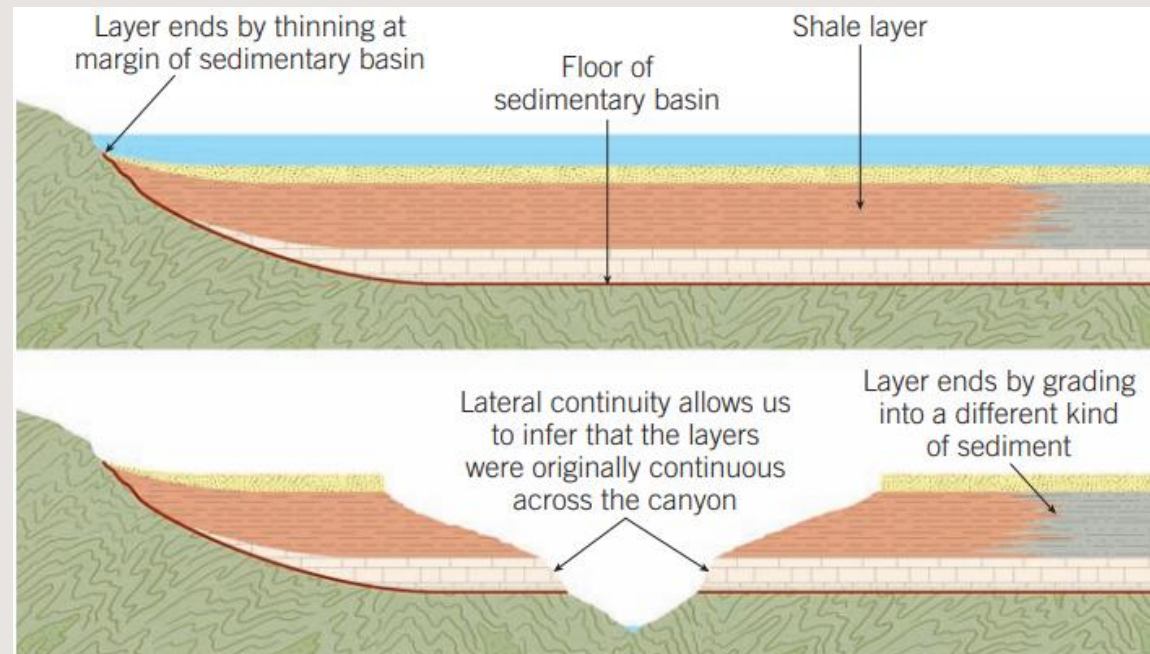


Basic Geologic Principles

3. *Principle of Lateral Continuity* - sedimentary beds originate as continuous layers that extend in all directions until they eventually grade into a different type of sediment or until they thin out at the edge of the basin of deposition.

Figure 4.3 Lateral continuity

Sediments are deposited over a large area in a continuous sheet. Sedimentary strata extend continuously in all directions until they thin out at the edge of a depositional basin or grade into a different type of sediment.



Basic Geologic Principles

4. *Principle of Cross-cutting Relationships* - states that geologic features that cut across rocks must form after the rocks they cut through.



Figure 4.4 Cross-cutting fault The rocks are older than the fault that displaced them. (Morley Read/Alamy)

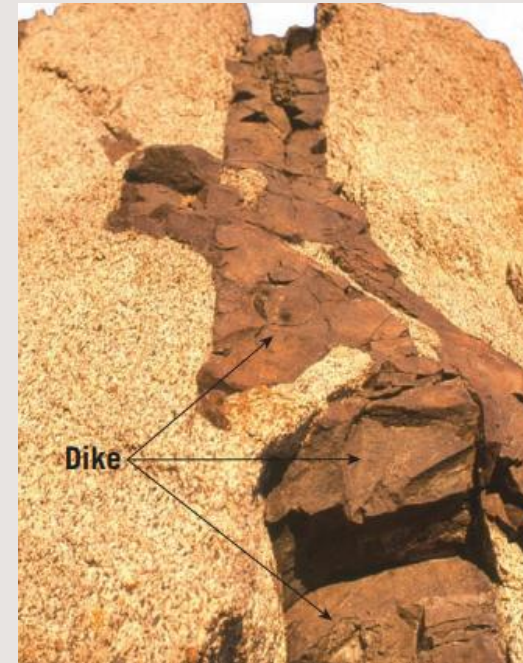


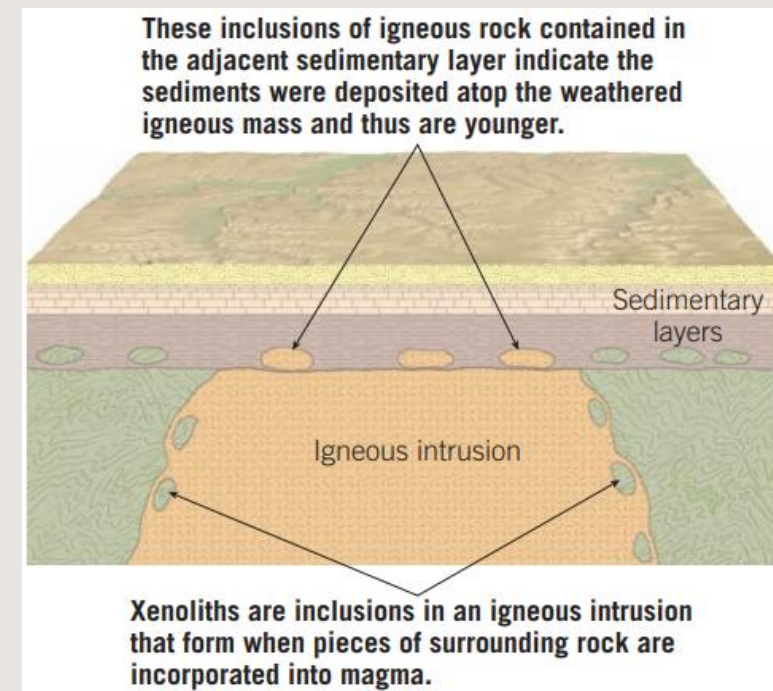
Figure 4.5 Cross-cutting dike An igneous intrusion is younger than the rocks that are intruded. (Photo by Ken M. Johns/Photo Researchers, Inc.)

Basic Geologic Principles

5. *Inclusions* - are fragments of one rock unit that have been enclosed within another. The rock mass adjacent to the one containing the inclusions must have been there first in order to provide the rock fragments. Therefore, the rock mass that contains inclusions is the younger of the two.

Figure 4.6 Inclusions

The rock containing inclusions is younger than the inclusions.



Basic Geologic Principles

6. *Unconformity* - represents a long period during which deposition ceased, erosion removed previously formed rocks, and then deposition resumed. In each case, uplift and erosion are followed by subsidence and renewed sedimentation.

Types of Unconformity:

- a. *Angular Unconformity* - consists of tilted or folded sedimentary rocks that are overlain by younger, more flat-lying strata. It indicates that during the pause in deposition, a period of deformation (folding or tilting) and erosion occurred.
- b. *Disconformity* - is a gap in the rock record that represents a period during which erosion rather than deposition is occurring.
- c. *Nonconformity* - younger sedimentary strata overlie older metamorphic or intrusive igneous rocks.

Basic Geologic Principles

6. *Unconformity*

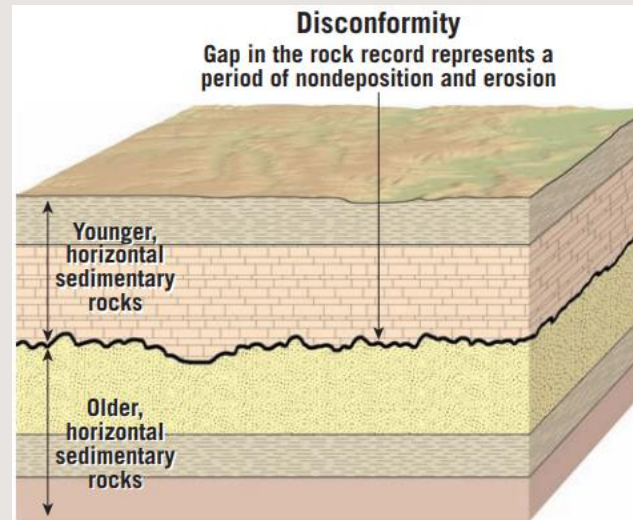
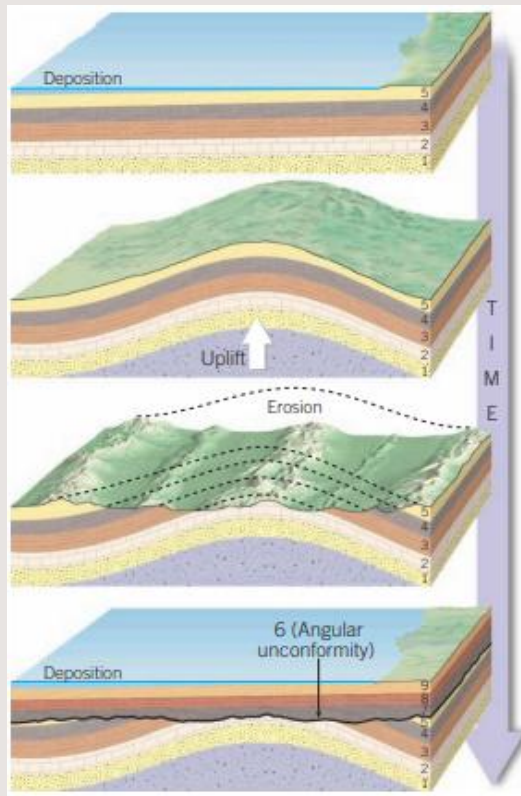


Figure 4.8 Disconformity The layers on both sides of this gap in the rock record are essentially parallel.

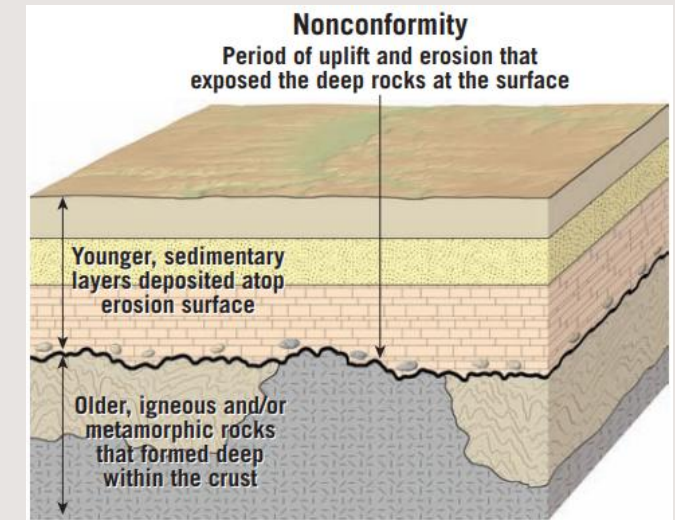


Figure 4.9 Nonconformity Younger sedimentary rocks rest atop older Metamorphic or igneous rocks.

Figure 4.7 Formation of an angular unconformity An angular unconformity represents an extended period during which deformation and erosion occurred.

Fossils: Evidence of Past Life

Fossils - the remains or traces of prehistoric life, are important inclusions in sediment and sedimentary rocks. They are basic and important tools for interpreting the geologic past. The scientific study of fossils is called [paleontology](#).

Types of Fossils:

1. *Permineralization* – a process which occurs when mineral-rich groundwater permeates porous tissue such as bone or wood, minerals precipitate out of solution and fill pores and empty spaces.
2. *Molds and Casts* – created when a shell or other structure is buried in sediment and then dissolved by underground water.
3. *Carbonization and Impressions* – Carbonization occurs when fine sediment encases the remains of an organism. If the film of carbon is lost from a fossil preserved in finegrained sediment, a replica of the surface, called an impression, may still show considerable detail.
4. *Amber* - the hardened resin of ancient trees.
5. *Trace Fossils* – Tracks, Burrows, Coprolite, Gastroliths

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Fossils

Figure 4.10 Types of Fossils

(Photos by **A**, Bernhard Edmaier/Photo Researchers, Inc.; Younger

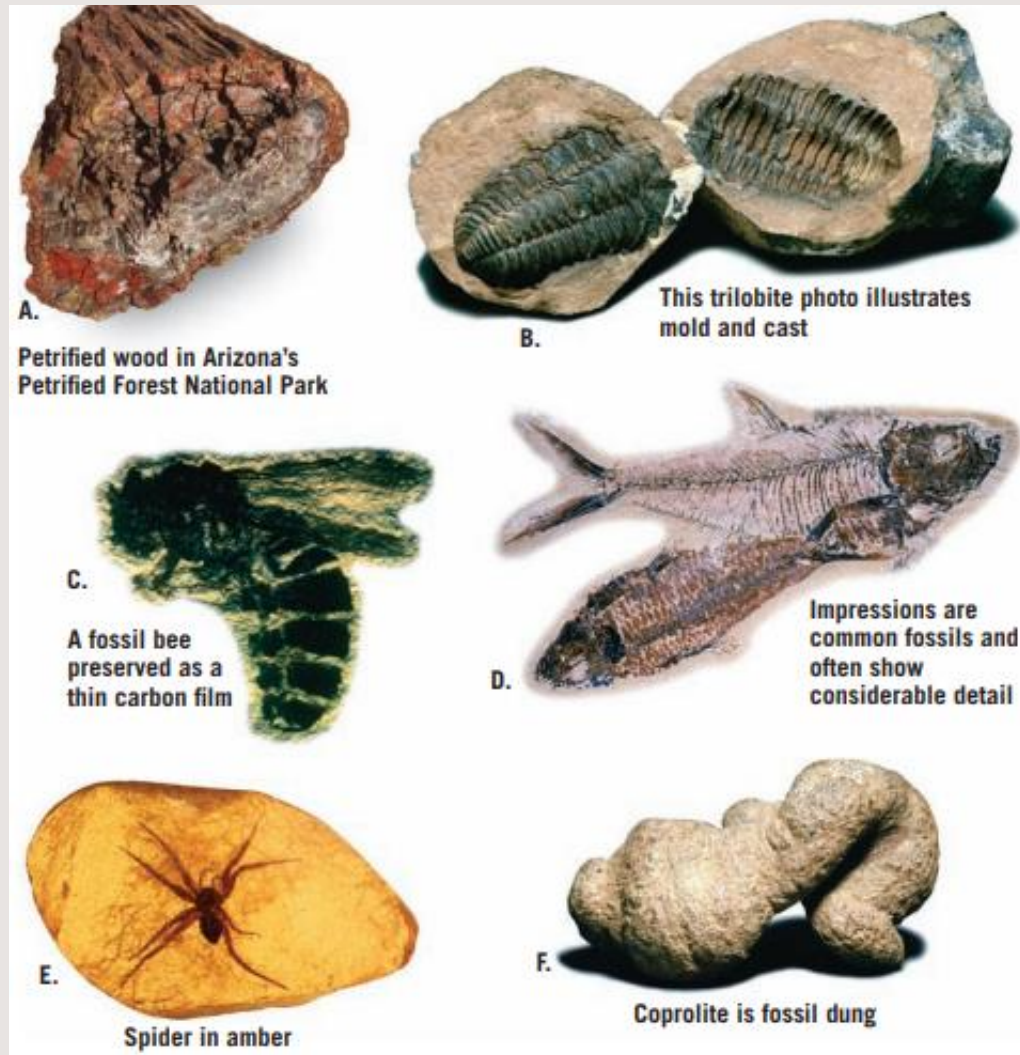
B, E. J. Tarbuck;

C, Florissant Fossil Beds National Monument;

D, E. J. Tarbuck;

E, Colin Keates/Dorling Kindersley Media Library

F, E. J. Tarbuck)



Absolute Dating

- Provides an approximate of the “numerical” date. It is also called the radiometric dating. Uses the principle of “half life” on the decay of radioactive parent material.

3 types of radioactive decays:

- a. **Alpha**-(α particles) may be emitted from the nucleus. An alpha particle is composed of 2 protons and 2 neutrons. Thus, the emission of an alpha particle means that the mass number of the isotope is reduced by 4, and the atomic number is lowered by 2
- b. **Beta**- When a beta particle (β particle), or an electron, is given off from a nucleus, the mass number remains unchanged because electrons have practically no mass. However, because the electron has come from a neutron (remember, a neutron is a combination of a proton and an electron), the nucleus contains one more proton than before. Therefore, the atomic number increases by 1
- c. **Electron capture**- Sometimes an electron is captured by the nucleus. The electron combines with a proton and forms a neutron. As in the last example, the mass number remains unchanged. However, because the nucleus now contains one less proton, the atomic number decreases by 1

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Absolute Dating

The following are the few common methods used in absolute dating:

1. Using different radioactive isotopes
2. Radiocarbon (carbon 14) dating

Radioactive Parent	Stable Daughter Product	Currently Accepted Half-Life Values
Uranium-238	Lead-206	4.5 billion years
Uranium-235	Lead-207	704 million years
Thorium-232	Lead-208	14.1 billion years
Rubidium-87	Strontium-87	47.0 billion years
Potassium-40	Argon-40	1.3 billion years

Table 4.1 Isotopes Frequently Used in Radiometric Dating

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Absolute Dating

2. Radiocarbon Dating – because the half-life of carbon-14 is only 5730 years, it can be used for dating events from the historic past as well as those from very recent geologic history. In some cases carbon-14 can be used to date events as far back as 70,000 years. Although carbon-14 is only useful in dating the last small fraction of geologic time, it has become a very valuable tool for anthropologists, archaeologists, and historians, as well as for geologists who study very recent Earth history.

The Geologic Time Scale

The geologic time scale subdivides the 4.6-billion-year history of Earth into many different units and provides a meaningful time frame within which the events of the geologic past are arranged.

EONS

ERAS

PERIODS

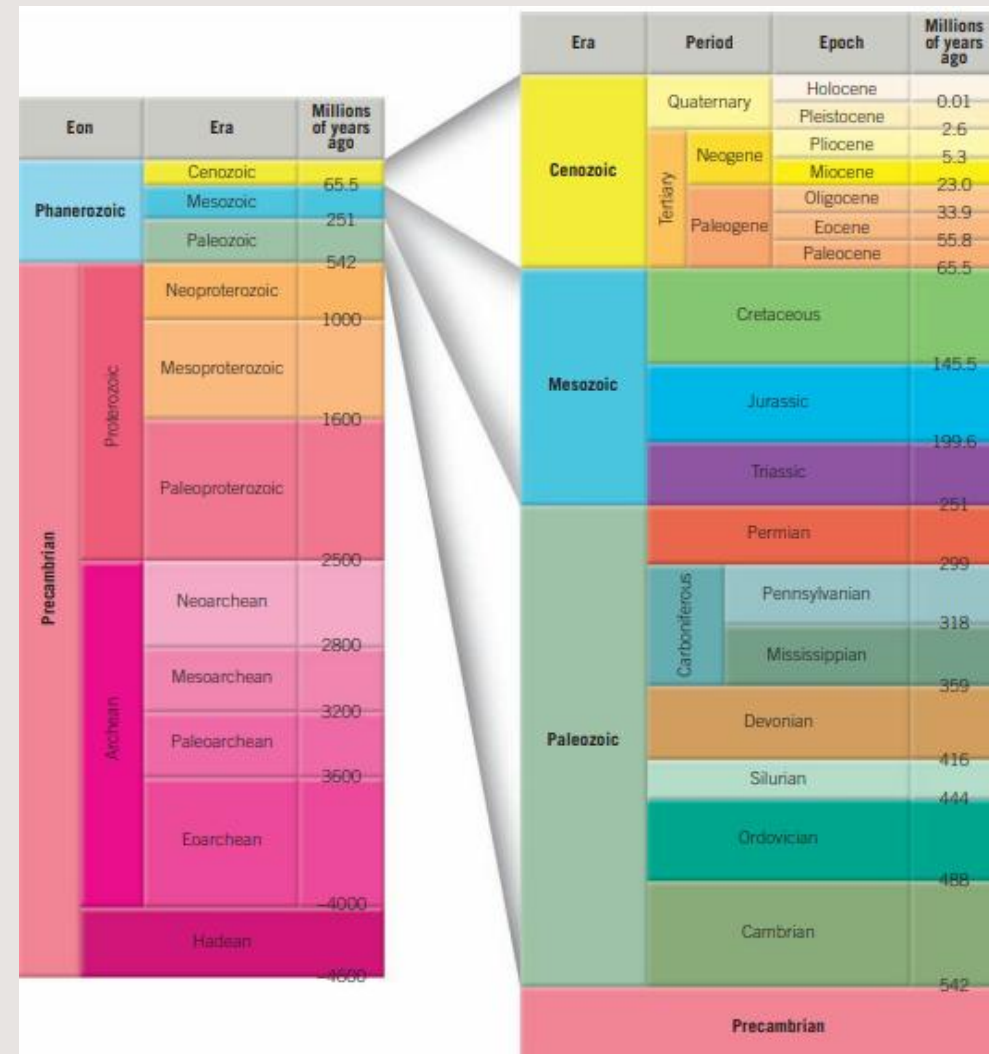
EPOCHS

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The Geologic Time Scale

Figure 4.11 Geologic time scale

Numbers on the time scale represent time in millions of years before present. Numerical dates were added long after the time scale was established using relative dating techniques.



Historical Geology

The Geologic Time Scale

Distribution of the current petroleum reserves

